



Write your student exam number in the boxes

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Northern Beaches Secondary College
Manly Campus

2022

HIGHER SCHOOL CERTIFICATE

Year 11 Examination

Mathematics Advanced

General

Instructions

- Reading time – 10 minutes
- Working time – 2.5 hours
- Write using black/blue pen
- NESA approved calculators may be used
- A reference sheet is provided with this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total Marks:

90

Section I – 10 marks (pages 2-4)

- Attempt Questions 1–10
- Allow about 15 minutes for this section
- Answer on Multiple Choice Answer Sheet provided

Section II – 80 marks (pages 5-22)

- Attempt Questions 11–33
- Allow about 2 hour and 15 minutes for this section
- Answer in the spaces provided

This paper MUST NOT be removed from the examination room

MARKS	MC	Q11-18	Q19-25	Q26-28	Q29-30	Q31-33	TOTAL
STUDENT MARK							
MAXIMUM	10	18	19	15	14	14	90



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Multiple-Choice Answer Sheet

Select the alternative A, B, C, or D that best answers the question by placing a **X** in the box.

	A	B	C	D
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Section I - 10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

1) Which of the following is equivalent to $(2x + y)^2 - (2x - y)^2$?

- A. 0 B. y^2 C. $4xy$ D. $8xy$

2) Which of the following is equal to $\log_2 5$?

A. $\ln 5 - \ln 2$

B. $\ln 5 + \ln 2$

C. $\frac{\ln 5}{\ln 2}$

D. $\frac{\ln 2}{\ln 5}$

3) Let f and g be two functions such that $f(3) = 6$, $f(5) = 3$, $g(6) = 3$, $g(4) = 5$.
What is the value of $f(g(4))$?

- A. 3 B. 4 C. 5 D. 6

4) Which of following is the natural domain of $y = \ln(2x - 1)$?

A. $\left[\frac{1}{2}, \infty \right)$

B. $\left[-\infty, \frac{1}{2} \right)$

C. $\left(\frac{1}{2}, \infty \right)$

D. $\left(-\infty, \frac{1}{2} \right)$

5) How many solutions does the equation $\sin^2 x + 2\sin x = 0$ have for $0 \leq x \leq 2\pi$?

- A. 1 B. 2 C. 3 D. 4

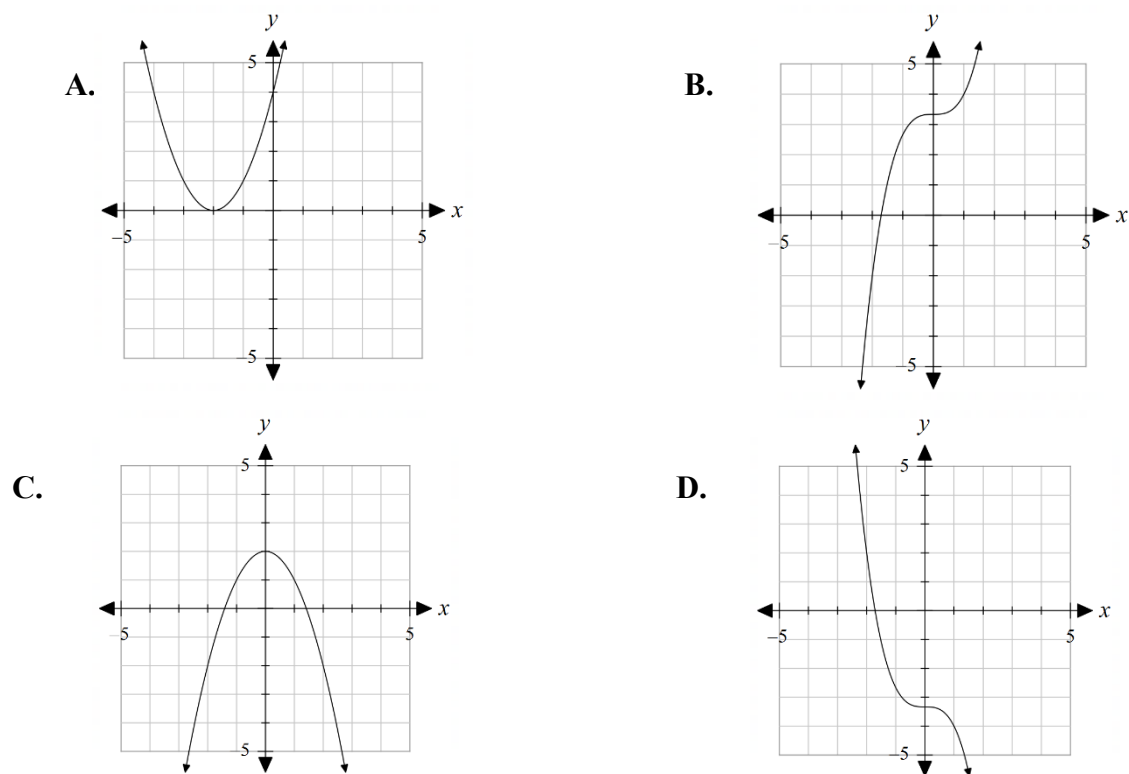
6) For the random variable X , the following probability distribution is given.

x	0	1	2	3
$P(X = x)$	$3k$	0.2	0.4	k

What is $P(X \geq 2)$?

- A. 0.2
B. 0.3
C. 0.4
D. 0.5

7) Which of the following graphs would have a tangent of $y = -2x + 3$ at the point where $x = 1$?



8) Which of the following is equivalent to $\frac{\cos 60^\circ}{1 - \cos(90^\circ - \theta)} + \frac{\cos 240^\circ}{1 - \sin(360^\circ - \theta)}$?

- A. $\sec^2 \theta$
- B. $\cot \theta \cos \theta$
- C. $\tan \theta \sec \theta$
- D. $\cot \theta \sec \theta$

9) A and B are two events such that $P(\overline{A}) = 0.3$, $P(B) = 0.5$ and $P(A \cap B) = 0.4$.
What is $P(A \cup B)$?

- A. 0.5 B. 0.6 C. 0.7 D. 0.8

10) If $y = xf(x^2)$ which of the following is $\frac{dy}{dx}$?

- A. $xf(x^2) + f(2x)$
- B. $f(x^2) + 2x^2 f'(x^2)$
- C. $f(x^2) + 2x^2 f(x^2)$
- D. $f(x^2) + x^2 f'(2x)$

2022

HIGHER SCHOOL CERTIFICATE

Year 11 Examination

Mathematics Advanced

Section II Answer Booklet

80 marks

Attempt Questions 11–33

Allow about 2 hour and 15 minutes for this section

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
 - Your responses should include relevant mathematical reasoning and/or calculations.
 - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
-

Question 11 (4 marks)

a) Simplify $\frac{1}{x-2} - \frac{4}{x^2-4}$

2

b) Simplify $\frac{(x^{-3}y^2)^3}{y^7} \times \frac{x^2y^{-3}}{xy^3}$ expressing your answer with positive indices.

2

Question 12 (2 marks)

Solve $|3x - 6| = 15$

2

Question 13 (2 marks)

Find 9^x if $\log_3 7 = x$.

2

Question 14 (2 marks)

State the domain and range for the function $y = 2^{-x} - 3$.

2

Question 15 (2 marks)

Solve $\tan x = -\sqrt{3}$ for $0^\circ \leq x \leq 360^\circ$.

2

Question 16 (2 marks)

Given $\log_a 4 = x$ and $\log_a 6 = y$, find an expression for $\log_a 96$ in terms of x and y .

2

Question 17 (2 marks)

For what value of k does the equation $x^2 - 3x + k = 0$ have equal roots?

2

Question 18 (2 marks)

Solve $\log_2 x - \log_2 4 = \log_2 \left(\frac{x-3}{2} \right)$.

2

Question 19 (2 marks)

Find the values of a and b for the function below given that $g(x)$ is continuous for all x .

2

$$g(x) = \begin{cases} x^2 - 4 & x < -1 \\ ax + b & -1 \leq x \leq 2 \\ 3x + 6 & x > 2 \end{cases}$$

Question 20 (2 marks)

Given $\tan \theta = -\frac{8}{5}$ for an obtuse angle θ , find the exact value of $\sin \theta$.

2

Question 21 (4 marks)

$\angle ABC$	$\angle BAC = 0.6$
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[illegible]

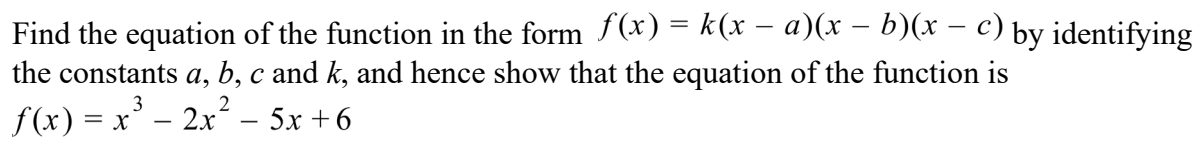
Question 22 (3 marks)

Find the equation of the tangent to the curve $y = x^2 e^{5x}$ at $x = 1$.

3

[illegible]

The graph of a cubic function $y = f(x)$ is shown below.



11

Question 24 (3 marks)

What is the average rate of change of the function

$$f(x) = \sin 2x \text{ between } x = \frac{\pi}{12} \text{ and } x = \frac{\pi}{8}?$$

Give your answer in exact form.

3

Question 25 (2 marks)

Three students sit a difficult English test. The probabilities that the three students pass the test are 25%, 40% and 55%. What is the probability that at least one student will pass?

2

Question 26 (7 marks)

Differentiate

a) $y = 3x^3 - 5x^2 + 2$

1

b) $y = \sqrt[3]{x} - \sqrt{x}$

1

c) $y = \frac{2+x^2}{x}$

2

d) $y = \frac{3x}{2x-1}$

1

e) $y = 5(4x+7)^3$

2

Question 27 (6 marks)

A biased die has a probability distribution for the outcome of the die being rolled as follows.

x	1	2	3	4	5	6
$P(X = x)$	0.2	0.2	0.1	0.25	0.1	0.15

a) Calculate the expected value.

1

b) Calculate $E(X^2)$ and hence find the variance.

2

c) Find $P(\mu - \sigma < X < \mu + \sigma)$.

2

d) Find $P(X \text{ is odd} \mid X > 3)$.

1

Question 28 (2 marks)

Sound intensity is measured in watts per square metre, W/m^2 . A sound that is just barely audible to the human ear is taken as a threshold intensity, $I_0 = 10^{-12} W/m^2$.

The loudness of a sound, B , in decibels (dB) is measured on a logarithmic scale and can be calculated with the formula $B = 10 \log_{10} \frac{I}{I_0}$

- a)** Show that the loudness of the threshold sound is 0 dB

1

Occupational health and safety regulations state that hearing protection must be worn if loudness levels in an area exceed 85 dB.

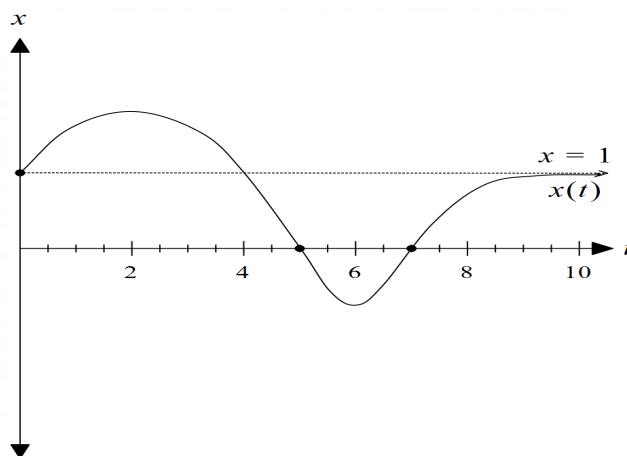
- b)** An industrial machine has a sound intensity of $0.05 W/m^2$. Determine if workers should be required to wear hearing protection when working near this machine.

1

Question 29 (5 marks)

A particle is moving along the x -axis, and its displacement from the origin is given by $x(t)$, where t is the time in seconds. $x(t)$ has a horizontal asymptote along the line $x = 1$.

The graph of $x(t)$ is shown below.



a) When is the particle at rest?

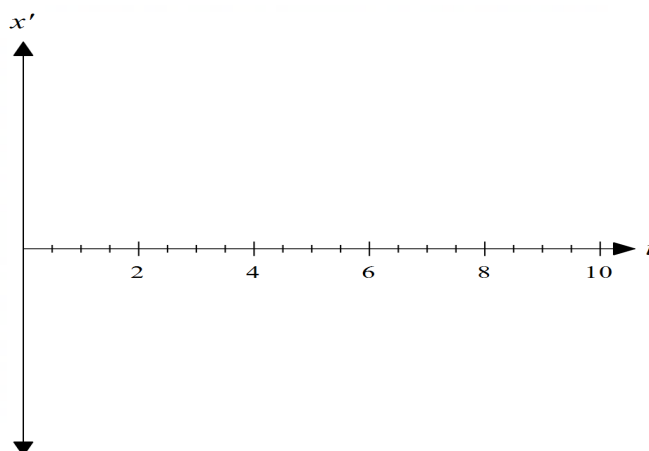
1

b) When is the particle to the left of the origin?

1

c) Sketch a graph of $x'(t)$. Mark any intercepts and asymptotes.

3



Question 30 (9 marks)

The koala population, P , in a sanctuary is modelled by the formula $P = 150 - 60e^{-0.02t}$, where t is time in weeks after the observations commence.

- a) Use differentiation to explain why the population is increasing.

2

- b) What is the initial koala population?

1

- c) What will the koala population be after 50 weeks?
Give your answer correct to the nearest whole number.

1

Question 30 continues on page 18

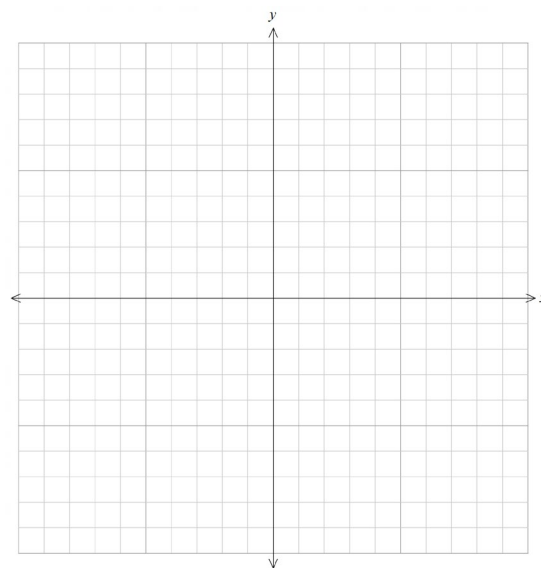
Question 30 (continued)

- d)** How long does it take for the population to reach 140?
Give your answer correct to the nearest week.

2

- e)** Sketch a graph of the model on the axis below.

2

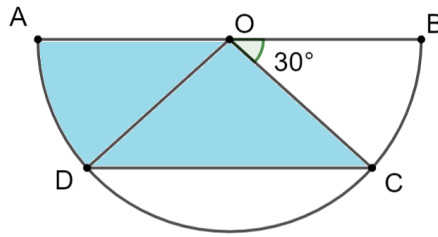


- f)** Explain why the population cannot exceed 150.

1

Question 31 (4 marks)

In the diagram, O is the centre of a semicircle, where $AB = 12$, $\angle BOC = 30^\circ$ and $AB \parallel CD$.



Find the exact area of the shaded region.

4

Question 32 (5 marks)

a) Prove the following identity: $\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta + 1} = \sec\theta$

2

b) Hence solve the equation $\frac{\sin 2\theta}{\cos 2\theta} + \frac{\cos 2\theta}{\sin 2\theta + 1} = 2$ for $0 \leq \theta \leq 2\pi$.
Give your answer(s) in radians.

3

Question 33 (5 marks)

In a group of 35 students, it is known that:

- 16 students study Biology
- 18 students study Chemistry
- 9 students study neither Biology nor Chemistry

- a) By using a Venn diagram, find the probability that a randomly selected student studies both Biology and Chemistry.

2

- b) Given that a randomly selected student studies Chemistry, what is the probability that they do NOT study Biology?

1

Question 33 continues on page 22

Question 33 (continued)

- c) Two students are randomly selected, one after the other.

Given that the first student studies Chemistry, what is the probability that the second student studies Biology?

2

End of paper

	Solutions	Marking Guidelines
1	$= ((2x + y) + (2x - y))((2x + y) - (2x - y))$ $= (4x)(2y)$ $= 8xy$	D
2	USE CHANGE OF BASE FORMULA ON REF SHEET	C
3	$f(g(4)) = f(5) = 3$	A
4		C
5	$\sin^2 x + 2\sin x = 0, 0 \leq x \leq 2\pi$ $\sin x(\sin x + 2) = 0$ $\sin x = 0, x = 0, \pi, 2\pi$ $\sin x = -2, \text{ no solution}$	C
6	$4k + 0.6 = 1$ $4k = 0.4$ $k = 0.1$ $P(X \geq 2)$ $= P(X = 2) + P(X = 3)$ $= 0.4 + 0.1$ $= 0.5$	D
7	From $y = -2x + 1$ the gradient is negative at $x=1$	C

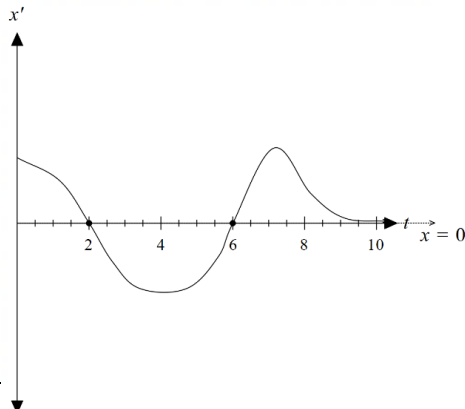
8	$\frac{\cos 60^\circ}{1 - \cos(90 - \theta)} + \frac{\cos 240^\circ}{1 - \sin(360 - \theta)}$ $= \frac{\cos 60^\circ}{1 - \sin\theta} + \frac{\cos 240^\circ}{1 - (-\sin\theta)}$ $= \frac{1}{2} \left(\frac{1}{1 - \sin\theta} \right) - \frac{1}{2} \left(\frac{1}{1 + \sin\theta} \right)$ $= \frac{1}{2} \left(\frac{1}{1 - \sin\theta} - \frac{1}{1 + \sin\theta} \right)$ $\frac{1}{2} \left(\frac{1 + \sin\theta - (1 - \sin\theta)}{(1 - \sin\theta)(1 + \sin\theta)} \right)$ $\frac{1}{2} \left(\frac{2\sin\theta}{1 - \sin^2\theta} \right)$ $\frac{1}{2} \left(\frac{2\sin\theta}{\cos^2\theta} \right) = \frac{(\sin\theta)}{(\cos\theta)} \left(\frac{1}{\cos\theta} \right)$ $\tan\theta \cdot \sec\theta$	C
9	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $= 0.7 + 0.5 - 0.4$ $= 0.8$	D
10	$u = x$ $u' = 1$ $v = f(x^2)$ $v' = 2x f'(x^2) \text{ (chain rule)}$ $\therefore y' = f(x^2) + 2x^2 f'(x^2)$	B
11a	$\frac{1}{x-2} - \frac{4}{(x+2)(x-2)}$ $= \frac{(x+2) - 4}{(x+2)(x-2)}$ $= \frac{x-2}{(x+2)(x-2)}$ $= \frac{1}{x+2}$ Students who wrote (x-2)(x²-4) as the common denominator are encouraged to always look for the lowest common denominator.	2 marks – correct solution 1 mark – writes with common denominator
11b	$\frac{(x^{-3}y^2)^3}{y^7} \times \frac{x^2y^{-3}}{xy^3} = \frac{x^{-9}y^6}{y^7} \times xy^{-6}$ $= \frac{x^{-8}}{y^7}$ $= \frac{1}{x^8y^7}$	2 marks – correct solution 1 mark – some correct simplification

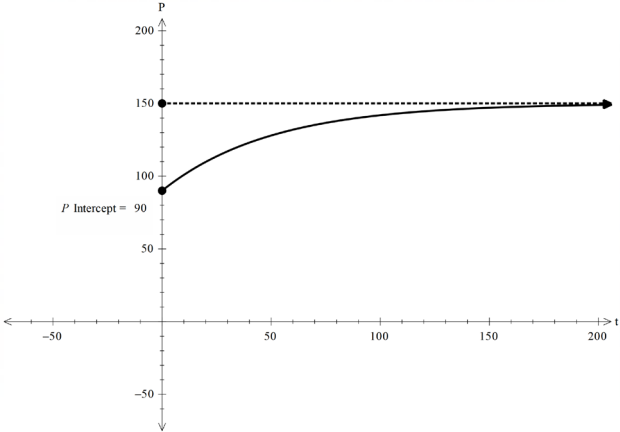
12	$3x - 6 = \pm 15$ $x = 7, x = -3$	2marks – correct solutions
13	$7 = 3^x$ $9^x = (3^2)^x = (3^x)^2 = 7^2 = 49$ Students who used the calculator and change of base law are required to demonstrate use of change of base for full marks.	2 marks – correct solution 1 mark writes $3^x=7$
14	Domain $(-\infty, \infty)$ Range $(-3, \infty)$	1 mark for each
15	$\tan x = -\sqrt{3}$ related angle $x = 60^\circ$ tan negative in Q 2 and Q 4 so $x = 120^\circ, 300^\circ$	2 marks – correct solution 1 mark – correct related angle but wrong quadrant(s)
16	$\log_a 96 = \log_a(4^2 \times 6)$ $= \log_a 4^2 + \log_a 6$ $= 2\log_a 4 + \log_a 6$ $= 2x + y$	2 marks – correct solution 1 mark – some correct use of log law
17	Equal roots $\Delta = 0$ $b^2 - 4ac = 0$ $(-3)^2 - 4 \times 1 \times k = 0$ $9 - 4k = 0$ $4k = 9$ $k = \frac{9}{4}$	2 marks – correct solution 1 mark – finds discriminant correctly, or solves incorrect discriminant=0
18	$\text{LHS} = \log_2\left(\frac{x}{4}\right)$ $\therefore \frac{x}{4} = \frac{x-3}{2}$ $\frac{x}{4} = \frac{2(x-3)}{4}$ $x = 2(x-3)$ $x = 2x - 6$ $x = 6$	2 marks – correct solution 1 mark – simplifies LHS using log law
19	For $x = -1$ $g(-1) = (-1)^2 - 4 = -3$ $-a + b = -3 \Rightarrow b = a - 3$ ① For $x = 2$ $g(2) = 3(2) + 6 = 12$ $2a + b = 12$ ② sub ① into ② $2a + a - 3 = 12 \Rightarrow a = 5$ sub $a = 5$ into ① $b = 2$	2 marks for correct solution 1 mark for finding $g(-1)$ and $g(2)$

	Alternative method: For $x = -1$, $g(-1) = (-1)^2 - 4 = -3 \therefore (-1, -3)$ For $x = 2$, $g(2) = 3(2) + 6 = 12 \therefore (2, 12)$ $m = \frac{12 - (-3)}{2 - (-1)} = \frac{15}{3} = 5$ $y + 3 = 5(x + 1) \Rightarrow y = 5x + 2$ $\therefore a = 5 \text{ and } b = 2$	
20	hyp = $\sqrt{8^2 + 5^2} = \sqrt{89}$ $\sin\theta = \frac{8}{\sqrt{89}}$ (obtuse angle is in second quadrant where sine is positive) <i>Marker's comments:</i> <i>Many did not explain why value was positive and received only 1 mark</i>	2 marks for correct solution 1 mark for finding hyp
21	$\frac{\sin \angle ACB}{5} = \frac{\sin 0.6}{4}$ $\angle ACB = \sin^{-1}\left(\frac{5\sin 0.6}{4}\right) = 0.783556\dots$ OR $\pi - \sin^{-1}\left(\frac{5\sin 0.6}{4}\right) = 2.35803\dots$ (ambiguous case) $\angle ACB = 2.35803\dots$ (given AB is longest side, $\angle ABC$ is obtuse) $\angle ABC = \pi - 0.6 - \left(\pi - \sin^{-1}\left(\frac{5\sin 0.6}{4}\right)\right) = 0.184 \text{ rad}$	4 marks for correct solution 3 marks for correct solution without consideration to ambiguous case (ans 1.758 rad) 2 marks for finding correct angACB 1 mark for writing sine rule with correct substitutions
21	Alternate: $4^2 = AC^2 + 5^2 - 2(AC)(5)\cos 0.6$ $AC^2 - 10\cos 0.6 AC + 9 = 0$ $AC = \frac{10\cos 0.6 \pm \sqrt{100\cos^2 0.6 - 4(9)}}{2}$ $AC = 6.960 \text{ or } 1.293$ But AB is longest side so $AC = 1.293$ $\frac{\sin \angle ABC}{1.293} = \frac{\sin 0.6}{4} \Rightarrow \angle ABC = 0.184$ OR $\cos \angle ABC = \frac{5^2 + 4^2 - 1.293^2}{2 \times 5 \times 4} \Rightarrow \angle ABC = 0.184$	4 marks for correct solution 3 marks for correct solution without noting the different cases 2 marks for correctly writing expression to find AC 1 mark for writing cosine

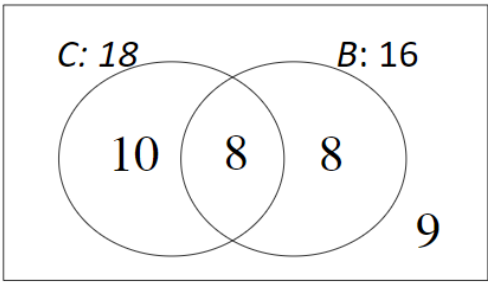
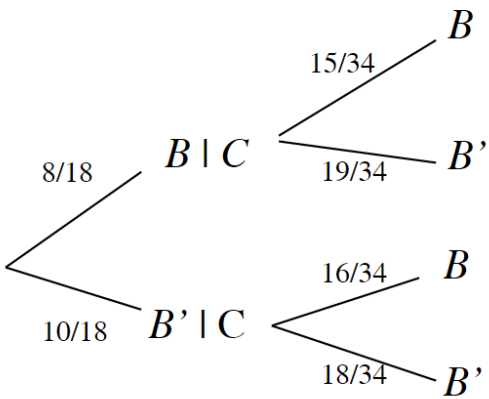
		rule with correct substitutions
21	<i>Marker's comments:</i> - Many students converted to degrees which made more work to solve question, and introduced potential for more errors in converting between radians and degrees - Conversions to degrees also resulted in less accurate final answers (not penalised as long as adequate working shown throughout process) - Error in finding missing length using Pythagoras' Theorem - Error in finding $\sin C = 0.7058$ versus angle C, and using that result in subsequent work	
22	$U = x^2 \qquad V = e^{5x}$ $U' = 2x \qquad V' = 5e^{5x}$ $\frac{dy}{dx} = 2xe^{5x} + 5x^2e^{5x}$ at $x = 1$, $\frac{dy}{dx} = 2e^5 + 5e^5 = 7e^5$ and $y = e^5$ $\therefore$ equation of tangent $y - e^5 = 7e^5(x - 1)$ $y = 7e^5x - 6e^5$	3 marks for correct solution 2 marks correctly finding gradient at $x=1$ and y value 1 mark for correctly differentiating
22	<i>Marker's comments:</i> Keep values as exact. i.e. students wrote $7e^5$ as 1038.89 (or full calculator display). Full marks still awarded in both cases.	
23	$f(x) = k(x+2)(x-1)(x-3)$ $(0,6) \quad 6 = k(2)(-1)(-3) \Rightarrow k = 1$ $\therefore f(x) = (x+2)(x-1)(x-3)$ $= (x^2 + x - 2)(x - 3)$ $= x^3 - 3x^2 + x^2 - 3x - 2x + 6$ $= x^3 - 2x^2 - 5x + 6$	3 marks for correct solution 2 marks for showing why $k=1$ 2 marks for using intercepts to find $f(x)$ and showing all steps to expansion 1 mark for using intercepts to write $f(x)$
24	$\frac{f\left(\frac{\pi}{8}\right) - f\left(\frac{\pi}{12}\right)}{\frac{\pi}{8} - \frac{\pi}{12}} = \frac{\sin\left(\frac{\pi}{4}\right) - \sin\left(\frac{\pi}{6}\right)}{\frac{\pi}{24}} = \frac{\frac{\sqrt{2}-1}{2}}{\frac{\pi}{24}} = \frac{12(\sqrt{2}-1)}{\pi}$	3 marks for correct solution 1 mark for finding values for

	<i>Marker's comments:</i> <i>Some students differentiated function and substituted into the derivative</i> <i>Some students were finding the average of $\sin(\pi/4)$ and $\sin(\pi/6)$</i>	$\sin(\pi/4)$ and $\sin(\pi/6)$ 1 mark for showing understanding on how to calculate average rate of change
25	$P(\text{at least one pass}) = P(\text{not none})$ $= 1 - P(0 \text{ passes})$ $= 1 - 0.75 \times 0.6 \times 0.45$ $= 0.7975 = \frac{319}{400}$	2 marks for correct solution 1 mark for writing expression on how to find probability (i.e 2nd line of sample answer)
26a	$\frac{dy}{dx} = 9x^2 - 10x$	1 mark correct solution
26b	$y = x^{\frac{1}{3}} - x^{\frac{1}{2}}$ $\frac{dy}{dx} = \frac{1}{3}x^{-\frac{2}{3}} - \frac{1}{2}x^{-\frac{1}{2}}$	1 mark correct solution
26c	$y = 2x^{-1} + x$ $\frac{dy}{dx} = -2x^{-2} + 1$	2 marks correct solution 1 mark partial correct with correct progress to index form
26d	$\frac{dy}{dx} = \frac{(2x-1)(3) - 3x(2)}{(2x-1)^2}$ $= -\frac{3}{(2x-1)^2}$	1 mark correct solution
26e	$\frac{dy}{dx} = 15(4x+7)^2 \times 4$ $= 60(4x+7)^2$	2 marks correct solution 1 mark partial correct use of $f'(f(x))$ rule with only one error
27a	$E(X) = 3.3$	1 mark correct solution
27b	$E(X^2) = 13.8$ $Var(X) = 13.8 - 3.3^2 = 2.91$	2 marks correct solution 1 mark partial correct with only

		one error in either part
27c	$\sigma = \sqrt{2.91} = 1.705\dots$ $\mu - \sigma \approx 1.59, \mu + \sigma \approx 5.006$ $P(\mu - \sigma < X < \mu + \sigma) = P(1.59 < X < 5.006)$ $= P(X = 2, 3, 4, 5)$ $= 0.65$	2 marks correct solution 1 mark partial correct finds bounds of prob.
27d	$\frac{P(X \text{ odd} \cap X > 3)}{P(X > 3)}$ $= \frac{P(X = 5)}{P(X = 4, 5, 6)}$ $= \frac{0.1}{0.5}$ $= \frac{1}{5}$	1 mark correct solution
28a	$B = 10 \log_{10} \frac{I_0}{I_0}$ $= 10 \log_{10} 1$ $= 10 \times 0$ $= 0$	1 mark correct solution
28b	$b = 10 \log_{10} \frac{0.05}{10^{-12}}$ $= 10 \times 10.69897$ $= 106.9897$ > 85 $\therefore$ workers MUST wear hearing protection	1 mark correct solution
29	(a) At rest where $dx/dt = 0 \rightarrow$ at $t = 2, 6$	1 Mark: Correct answer
	(b) Left of $x = 0$ when graph is below the t axis $\rightarrow 5 < t < 7$	1 Mark: Correct answer
	(c) 	3 Marks: Correct answer 2 Marks: 1 mistake 1 Mark: Correct intercepts or asymptote

30a	$P = 150 - 60e^{-0.02t} \rightarrow \frac{dP}{dt} = 0 - 60e^{-0.02t} \cdot -0.02$ $= 1.2e^{-0.02t}$ $> 0 \text{ since } e^{-0.02t} > 0 \text{ for all } t$ $\therefore P \text{ is increasing.}$	2 Marks: Correct answer 1 Mark: Obtains a correct derivative OR Obtains a value from an incorrect derivative of equal merit
b	$t = 0 \rightarrow P = 150 - 60e^0 = 90$	1 Mark: Correct answer
c	$t = 50 \rightarrow P = 150 - 60e^{(-0.02) \cdot 50} = 127.927... = 128$	1 Mark: Correct answer
d	$P = 150 - 60e^{-0.02t}$ $P = 140 \rightarrow 140 = 150 - 60e^{-0.02t}$ $-10 = -60e^{-0.02t} \rightarrow \frac{1}{6} = e^{-0.02t}$ $-0.02t = \ln\left(\frac{1}{6}\right) \rightarrow t = \frac{\ln\left(\frac{1}{6}\right)}{-0.02}$ $t = 89.587... \text{ weeks}$ $t \approx 90 \text{ weeks}$	2 Marks: Correct answer 1 Mark: Obtains $e^{-0.02t} = \frac{1}{6}$
e	 The graph shows a coordinate plane with P on the vertical axis and t on the horizontal axis. The vertical axis has labels at 50, 100, 150, and 200. The horizontal axis has labels at -50, 50, 100, 150, and 200. A curve starts at a point labeled 'P Intercept = 90' on the P-axis and increases, approaching a horizontal dashed line at P = 150 as t increases.	2 Marks: Correct answer 1 Mark: Obtains any two of correct P intercept, correct asymptote or concavity.
f	As $t \rightarrow \infty, e^{-0.02t} \rightarrow 0$ $\therefore P \rightarrow 150 - 0 = 150$ Since $e^{-0.02t} > 0$ for all t, P will always be less than 150.	1 Mark: Correct answer

31	$OA = OB = OC = OD = 6$ (radii) $\angle OCD = 30$ (alternate, $AB \parallel CD$) $\angle ODC = 30$ (base angles of isosceles Δ) $\angle DOC = 120$ (angle sum Δ) $\angle AOD = 30$ (straight angle) Area of ΔODC : $A = \frac{1}{2} (6)(6) \sin 120$ $A = 18 \times \frac{\sqrt{3}}{2} = 9\sqrt{3}$ Area of sector AOD: $A = \frac{30}{360} \times \pi(6)^2$ $A = 3\pi$ $\therefore \text{Area shaded} = 9\sqrt{3} + 3\pi$	4 marks 3 marks: finds the area of the triangle or the sector including the correct formulation of the total area 2 marks: provides correct formulation of the total area 1 mark: finds angle DOC or OD or OC
32a	$\begin{aligned} LHS &= \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta + 1} \\ &= \frac{\sin \theta(\sin \theta + 1) + \cos^2 \theta}{\cos \theta(\sin \theta + 1)} \\ &= \frac{\sin^2 \theta + \sin \theta + \cos^2 \theta}{\cos \theta(\sin \theta + 1)} \\ &= \frac{1 + \sin \theta}{\cos \theta(\sin \theta + 1)} \\ &= \frac{1}{\cos \theta} \\ &= \sec \theta \end{aligned}$	2 marks 1 mark: correctly forms one fraction
32b	$\frac{\sin 2\theta}{\cos 2\theta} + \frac{\cos 2\theta}{\sin 2\theta + 1} = 2$ $\sec 2\theta = 2$ $\cos 2\theta = \frac{1}{2}$ $2\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$ $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$	3 marks 1 st mark: uses part (a) successfully 2 nd mark: finds at least one solution to the equation with working

33a	<i>Total: 35</i>  $35 - 9 = 26$ students study at least one $18 + 16 - 26 = 8$ students study both Therefore, $8/35$	2 marks 1 mark for finding the number of students who study both Bio and Chem, or equivalent merit
33b	$10/18 = 5/9$	1 mark
33c	There are two cases: the first student studies Biology, or the first student does not study Biology.  $P(\text{2nd Bio}) = \frac{8}{18} \times \frac{15}{34} + \frac{10}{18} \times \frac{16}{34} = \frac{70}{153}$	2 marks 1 mark for some consideration of two cases / consideration of 'without replacement'
34		
35		